

Beyond Deep Impostors

Improving the Deep Impostors approach with LLMs

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What This Talk Covers

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Motivation & Foundation

The authorship question and the Deep Impostors method

02

Our Approach

Rewriting plays toward the majority corpus style in big chunks

03

Method

Three distance metrics and their limits

04

Implementation

Majority-style extraction, the pipeline, and the DistilBERT extension

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Evaluation & Outlook

Success criteria, expected behaviour, and Phase B

“

*From a centuries-old
literary puzzle to a single,
measurable stylistic signal.*

Research track · Phase A

The Shakespeare Authorship Question



A debate measured in statistics, not opinion

For over two centuries, scholars have debated whether the works attributed to William Shakespeare were written by him, by a single contemporary, or by a collaborative network of Elizabethan authors.

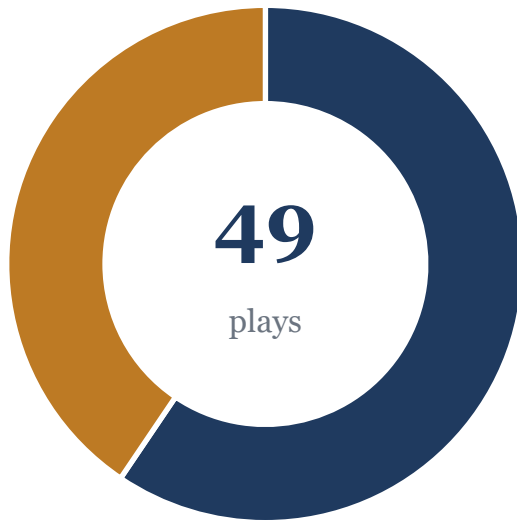
Computational stylometry attacks this question by measuring authorial “fingerprints” the subtle statistical regularities a writer produces involuntarily.



The bottleneck

Verification needs **impostor texts**, genuine works by other authors that are contemporaneous, comparable in length and topic, and confidently attributed.

Deep Impostors (Avros & Volkovich, 2025)



34 classified as authentic Shakespeare

CNN/BI-LSTM 15 surfaced as a stylistically inconsistent cluster



Headline result: the pipeline surfaces a cluster of **15 plays** whose stylistic signature is inconsistent with the rest of the canon.

Rewriting Toward the Majority Corpus Style



**Can LLMs improve the Deep Impostors idea?
Rewrite every play in the majority style and
measure what moves**

We extract the majority corpus style and an LLM rewrites each play toward it in big, near-play-sized chunks. The plot stays, the style shifts, so the original-vs-rewrite distance becomes the authorship signal.

Original play



LLM rewrite (same plot, majority style)

Research Goal

Test whether LLMs can improve the Deep Impostors framework. Using three complementary distance metrics, one classical and two neural, we compare each original play with its majority-style rewrite: a large distance means the original was not written in the majority corpus style, marking that play as an outlier.

Why Rewriting, Not Generation



01

Content held constant

Rewriting preserves plot, characters, and structure, so semantic content stays roughly fixed.



02

Style is the variable

Only the stylistic axis moves — precisely what the three distance metrics are built to measure.



03

Conditioned on a target

The rewrite is steered toward an extracted corpus-majority-style profile, not a vague “modern English.”

Holding meaning fixed while moving style is what isolates the stylistic axis our metrics operate on.

Three Distance Metrics: Classical + Neural

Style Distance

$$d = 1 - \cos(SD(A), SD(B))$$

Neural style embedding, content-independent

LLM-as-Judge

$$d = \text{mean}(s_1 \dots s_6) / 4$$

Reasoned style shift over 6 dimensions

Burrows's Delta

$$\Delta = (1/n) \sum | z_i(A) - z_i(B) |$$

Function-word style, classical anchor

Every Metric Has a Blind Spot

METRIC	KEY WEAKNESS
Style Distance	Trained on modern text; may drift on archaic verse
LLM-as-Judge	Less reproducible; guard against training-data contamination
Burrows's Delta	Reference vocabulary shifts when register changes

No single ruler is enough

Two neural views capture what word-counts miss; Delta stays interpretable. We trust all three agreeing.

What Did Not Work: Small-Chunk Rewriting



First attempt: split each play into small segments and rewrite each one independently.

Lost dramatic context

Chunks stripped of speaker labels and situational frame;
character voices collapsed into a flat, homogenised register.

Boundary artefacts

Truncated end-of-chunk sentences and cross-chunk repetition
skewed the token and n-gram statistics.

Both effects produced distance distributions whose noise drowned the signal we cared about.



Rewriting in Big, Near-Play-Sized Chunks

The Rewrite Pipeline in Action

Majority-style directive (LLM-extracted): elevated blank verse, heavy figurative imagery, frequent inversion, iambic cadence, archaic function words (thou, hath, ere).

Original passage (flagged as outlier)

I have to go up to the tower now
and tell them the king was killed tonight.



Rewritten toward majority (canonical) style

*Now to the tower must I haste, to tell
How our dread sovereign this night was slain.*

StyleDistance 0.71 • **Burrows Δ 0.55** • **LLM-judge 0.62**

Illustrative pair. In our runs, Arden of Feversham shows the largest gap (semantic distance ≈ 0.84), consistent with its disputed authorship; a play already in the majority style scores near 0.

Extracting the Majority Style (LLM)

Sampled from undisputed plays only. Output frozen and reused corpus-wide.

You are a descriptive stylometric analyst. Below are N passages sampled from the STYLISTIC CORE of the corpus. Characterize the COMMON, RECURRING writing style shared across them.

Rules:

- Describe ONLY what is present in the passages. Use no outside knowledge about any author, period, or work.
- Report what is TYPICAL across the set. If one passage differs, list it under variation, not the core profile.
- Be concrete and quantitative where you can.

Describe the shared style on six dimensions:

- | | |
|-------------------------|------------------------|
| 1 Sentence architecture | 2 Diction and register |
| 3 Syntactic complexity | 4 Figurative density |
| 5 Rhythm and meter | 6 Grammatical habits |

Return ONLY JSON:

```
{ "core_style": { ...one field per dimension... },  
  "observed_variation": [ ],  
  "rewrite_directive": "one paragraph telling a rewriter how to  
    move any passage toward this style WITHOUT changing meaning" }
```

Passages: <<< {sampled_core_passages} >>>

Rewriting Toward the Majority Style

Low temperature. Meaning and length preserved. Exemplars carry a small model.

You are an expert literary stylist. Rewrite the PASSAGE so its WRITING STYLE matches the TARGET as fully as possible, while keeping its MEANING and LENGTH unchanged. Change HOW it is said, never WHAT is said.

Rules:

- 1 Preserve meaning: keep every event, image, and name, in order.
Do not add, drop, summarize, or reinterpret content.
- 2 Converge fully to the target on all six style dimensions (architecture, diction, syntax, figures, rhythm, grammar).
A passage already in the target barely changes; a far one changes greatly; both END in the same style.
- 3 Preserve length within ~10% and keep paragraph / line breaks.
- 4 The EXAMPLES show STYLE only; never reuse their words or content.
- 5 Aim for the specific target, not generic modern or "old" English.
- 6 Output ONLY the rewrite, between the markers.

TARGET DIRECTIVE: <<< {rewrite_directive} >>>

TARGET EXAMPLES (style only): <<< {exemplar_1..3} >>>

PASSAGE: <<< {passage} >>>

REWRITE: <REWRITE> ... </REWRITE>

Scoring the Stylistic Shift (LLM Judge)

Blind A/B, per-dimension rubric, JSON output, averaged over chunk pairs.

You are a forensic stylometric analyst. You will see two passages, A and B, with similar content. Judge ONLY how different their WRITING STYLE is. Ignore plot, meaning, and subject. Use no outside knowledge about who wrote either passage or which is older.

Rate each dimension 0-4:

0 identical . 1 slight . 2 noticeable . 3 strong . 4 radical

Dimensions:

1 Sentence architecture	2 Diction and register
3 Syntactic complexity	4 Figurative density
5 Rhythm and meter	6 Grammatical habits

Return ONLY JSON:

```
{ "sentence_architecture": {"score":0,"note":""},  
  ... one entry per dimension ... ,  
  "overall_style_shift": 0.0, "summary": "" }
```

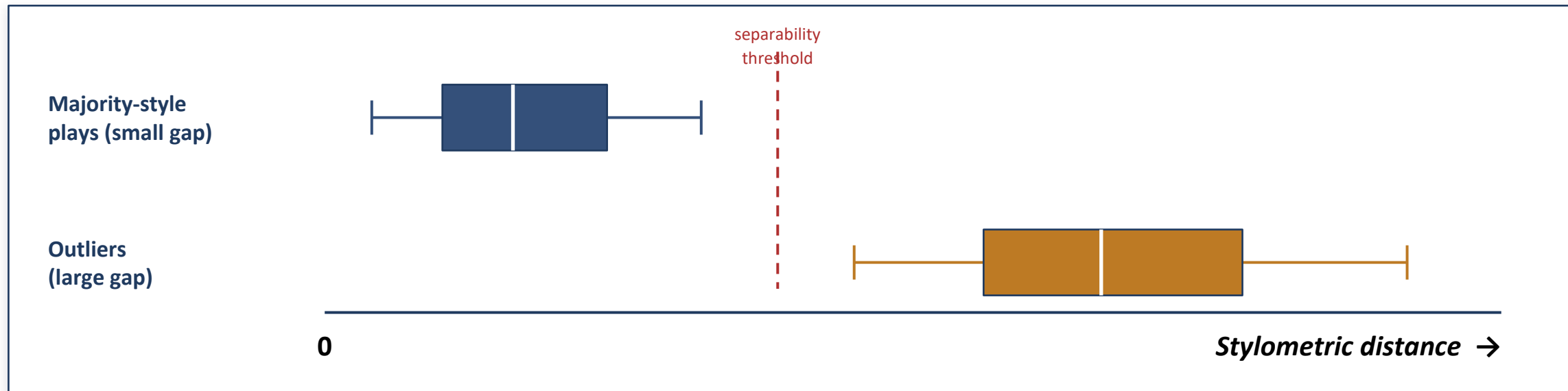
Distance = mean(scores) / 4. Randomize which text is A vs B, and average over several chunk pairs per play.

Passage A: <<< {passage_a} >>>

Passage B: <<< {passage_b} >>>

What We Expect to Observe

A play already written in the majority corpus style barely moves when rewritten, so its distance stays small. A play in a different style is pulled toward the majority, so its distance is large, flagging it as an outlier.



Schematic of the working hypothesis, not measured results. Phase B quantifies the gap per metric.

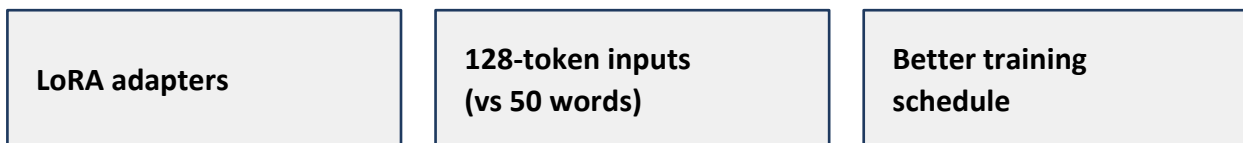
The DistilBERT Extension

A parallel improvement to the original pipeline: replace under-trained BERT separator with **DistilBERT**.

Reason:

Weak, by the authors' own admission, due to insufficient fine-tuning.

Three fixes:



How it works:

Fine-tune DistilBERT per impostor pair → train model → run every play through it → group the plays into two groups.

DistilBERT is a parallel track to the rewriting study; a full impostor-pair sweep is planned for Phase B.

The Mistral 7B Extension

The most powerful separator. (CNN → DistilBERT → **Mistral 7B**) progression. A 7-billion-parameter open-weight LLM, fine-tuned on impostor pairs with 4-bit QLoRA, replaces the paper’s neural separator.

Reason:

A generative LLM understands language far more deeply than a CNN or DistilBERT. We test whether that depth makes a sharper separator, and it benchmarks the LLM-as-separator idea behind the Phase B “LLM at every stage” extension.

How it works:

take two impostors

train a model to tell their
styles apart

run every play to get a
score

repeat for many
impostors

compare all the scores

group into two clusters

Preliminary result: GPU limits capped the run at 22 of 30 pairs. The downstream pipeline is identical to Deep Impostors; the full sweep is planned for Phase B.

Phase B: Two Extensions



1

Chunk-size sensitivity

Vary chunk size, from acts to near-full plays, to confirm the signal is stylistic and not a chunking artefact.



2

LLM at every stage

Replace separator, DTW, Isolation Forest and K-medoids with prompted LLMs; benchmark against the classical stack.

Each extension ships with example LLM prompts for the substituted stage: segment classifier, sequence comparator, anomaly detector, clusterer.

■ SUMMARY

Contribution of Phase A

An LLM-based improvement of the Deep Impostors framework: majority-style rewriting of the Shakespeare canon in big chunks, a three-metric distance harness for outlier detection, an explicit analysis of each metric's blind spots, and a defined Phase B research programme.

KEY REFERENCES

- Volkovich & Avros (2025) — Deep Impostors approach, Digital Scholarship in the Humanities
- Burrows (2002) — 'Delta': a measure of stylistic difference
- Devlin et al. (2019) — BERT; Brown et al. (2020) — GPT-3 / LLMs

Thank you · Questions welcome